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What Project Axiom Actually Solves

Seven real failure modes in enterprise Java infrastructure — and the exact, zero-dependency module that removes each one at the root.

[com.ensblu — live on Maven Central →](#)

Swipe →

The ORM Trap

THE PROBLEM

Heavy ORMs map relational tables to mutable object graphs — causing N+1 query explosions, dirty-checking overhead, and total loss of control over generated SQL.

THE FIX

Relational data treated as raw streams and flat immutable maps — persistence mapped directly to data structures, no object virtualization layer.

Solved by `axiom-warp-jdbc` — direct-to-PostgreSQL parameter binding, zero reflection.

The Null Reference Trap

THE PROBLEM

Unchecked `null` as a default missing-value indicator leads to scattered defensive checks and production-crashing `NullPointerException`s.

THE FIX

Absence becomes an explicit, typed data construct — algebraic types and sentinel structures instead of an implicit, uninitialized trap.

Solved by **axiom-language** — explicit Result and sum types at every boundary.

The Annotation & DI Container Magic

THE PROBLEM

Runtime classpath scanning and annotation-driven wiring hide object lifecycles, slow startup, and produce wiring failures only discoverable at runtime.

THE FIX

Explicit, manual composition — every dependency passed as a plain parameter, fully visible, with instant startup.

Solved by **axiom-sovereign** — the zero-reflection, zero-scanning core all other modules build on.

The Custom Parser Void

THE PROBLEM

Reflection-based JSON parsers (Jackson, Gson) blindly slurp payloads into mutable object graphs, throwing cryptic runtime exceptions on schema drift.

THE FIX

Hand-crafted parsers inspect raw byte streams directly, validating and transforming input into immutable maps and vectors — no reflection.

Solved by **axiom-spec** — explicit schema contracts validated at the perimeter, not inferred at runtime.

The Java Collections Desert

THE PROBLEM

Default reliance on HashMap/ArrayList means suboptimal memory layout, synchronization bottlenecks, and no true structural sharing under concurrency.

THE FIX

Persistent, immutable structures — Hash Array Mapped Tries and Vector Bitmapped Tries — for safe, thread-free, cache-friendly state.

Solved by **axiom-sovereign** — HAMT / VBT core data structures.

The Exception-Driven Control Flow

THE PROBLEM

Using exceptions for expected outcomes like "UserNotFound" costs stack-trace generation overhead and hides control flow inside try/catch jumps.

THE FIX

Expected outcomes return explicit Result containers and sum types — exceptions reserved strictly for unrecoverable faults.

Solved by **axiom-language** — Result-pattern return types throughout.

The Infinite Tooling Churn

THE PROBLEM

Constant upgrades to build plugins, annotation processors, and codegen tools that break compatibility on every minor release.

THE FIX

Zero-dependency foundations on raw JDK primitives (Loom, Amber) and standard language features that don't break underneath you.

Solved by `com.ensemblu` — all 6 modules, zero external dependencies, published on Maven Central.

ENSEMBLU

Worth a 10-minute technical look?

Every module above is live and pullable today — no sales process, just `com.ensemblu` on Maven Central.

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